

**Implementation of Clam Anti-Virus with Real Time
Protection, Automatic Quarantine, Logging on Ubuntu**

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Abstract

This project implements **Clam AV**, an open-source antivirus solution, to detect and prevent malware on a Linux system. The objective is to develop a simple, automated, and cost-effective endpoint security solution capable of identifying malicious files, providing real-time protection, and reducing the risk of malware infections.

The project involves installing and configuring Clam AV, **Fresh Clam** for automatic virus signature updates, and the **Clam AV Daemon (clam)** for efficient scanning. A custom real-time monitoring script continuously monitors selected directories and automatically scans newly created or modified files. If malware is detected, the infected file is quarantined, the event is logged. The monitoring script is managed using **systemd**, ensuring it starts automatically during system boot and runs continuously.

The system is tested using sample malware files to verify its ability to detect, quarantine, and report threats effectively. This project demonstrates practical concepts of malware detection, automated incident response, log management, and Linux system administration, providing a lightweight and reliable antivirus solution for personal systems, educational environments, and small organizations.

Introduction

Malware, short for malicious software, is any program designed to damage, disrupt, or gain unauthorized access to a computer system. As cyberattacks continue to increase, protecting computer systems from malware has become an essential part of cybersecurity. These threats can steal sensitive information, corrupt important files, slow down system performance, or even make a computer unusable.

Antivirus software plays an important role in detecting and removing malicious files before they can cause harm. Clam AV (Clam Antivirus) is a free and open-source antivirus software designed to detect and remove malware from computer systems. It is open source, anyone can use, modify, and improve the software, making it a popular choice for individuals, educational institutions, and organizations looking for a reliable and cost-effective antivirus solution.

Clam AV uses a signature-based detection method, where scanned files are compared against a regularly updated database of known malware signatures. The virus database is automatically updated using Fresh Clam, ensuring the antivirus can recognize newly discovered threats. For improved performance, Clam AV Daemon (clamd) runs as a background service, allowing faster and more efficient scanning.

Objective

The objectives of this project are:

- Install Clam AV on Linux environment.
- Configure automatic virus signature updates.
- Enable real-time file monitoring.
- Detect malware automatically.
- Prevent malware execution through quarantine.
- Generate detection logs.
- Automatically start protection after system boot.

LAB DEMONSTRATION

System Requirements

- Ubuntu Linux Server 24.02 LTS
- Clam AV Latest
- Clam AV Daemon latest
- Fresh Clam, ClamTK Latest
- Inotify-tools
- Libnotify-bin
- Systemd

LAB DEMONSTRATION

Clam AV's Working Principle

ClamAV performs the antivirus scanning on files as they are searched in the database of known malicious data and programs. For this implementation, the software keeps an eye on all specific user directories for any changes.

As soon as a changed or new file is identified, the real-time protection service calls the clamav daemon to scan the suspicious data for malicious content. In case of any malware detection, the infected file is moved to a specific quarantine folder to prevent its execution.

Additionally, the clamav daemon updates the log file, and the system journal is updated.

Installation and Configuration

- Update the Linux environment
- Sudo apt update
- Sudo apt upgrade -y

Installing the Packages

- sudo apt install \
Clamav \
Clamav-daemon \
Clamav-freshclam \
Clamtk \
Inotify-tools \
Libnotify-bin

Updating Virus Database

- First, stop the Fresh Clam service
sudo systemctl stop clamav-freshclam
- Update the database
sudo freshclam
- Enabling Freshclam
sudo systemctl enable clamav-freshclam
sudo systemctl start clamav-freshclam

Enable ClamAV Daemon

- Enable the Daemon service.
sudo systemctl enable clamav-daemon
- Start the Daemon service.
sudo systemctl restart clamav-daemon
- To verify the status,
sudo systemctl status clamav-daemon

Creating Quarantine folder

- creating a quarantine folder.
sudo mkdir -p /var/quarantine
sudo chmod 700 /var/quarantine

Creating Log Files

- Creating the log files
sudo touch /var/log/clamav/realtime.log
sudo chmod 664 /var/log/clamav/realtime.log
sudo chown root: clamav /var/log/clamav/realtime.log

Creating Real-Time Monitoring Script

- Paste the below script into the file, save the file using ctrl + O, exit from the editor.
#!/bin/bash

```
WATCH_DIR="/home"  
LOGFILE="/var/log/clamav/realtime.log"  
QUARANTINE="/var/quarantine"
```

```
mkdir -p "$QUARANTINE"
```

```
inotifywait -mr \  
-e create \  
-e modify \  
-e moved_to \  
--format '%w%f' \  
"$WATCH_DIR" |
```

```
while read FILE  
do  
    # Skip if it's not a regular file  
    [ -f "$FILE" ] || continue  
  
    echo "Scanning: $FILE"  
  
    RESULT=$(clamscan \  
        --fdpass \  
        --move="$QUARANTINE" \  
        --log="$LOGFILE" \  
        "$FILE")
```

```
    echo "$RESULT"
```

```
if echo "$RESULT" | grep -q "FOUND"  
then
```



```
notify-send \  
"⚠ Malware Detected" \  
"Infected file quarantined"
```

```
logger "ClamAV: Malware detected in $FILE"
```

```
fi
```

```
done
```

- Make the file executable
`sudo chmod +x /usr/local/bin/clamav-realtime.sh`

Create a systemd service

```
sudo nano /etc/systemd/system/clamav-realtime.service
```

- Paste the below script in the file, save it using `ctrl + O`, exit from the editor

```
[Unit]
```

```
Description=ClamAV Real-Time Protection
```

```
After=network.target clamav-daemon.service
```

```
[Service]
```

```
Type=simple
```

```
ExecStart=/usr/local/bin/clamav-realtime.sh
```

```
Restart=always
```

```
RestartSec=5
```

```
User=root
```

```
Group=root
```

```
StandardOutput=journal
```

```
StandardError=journal
```

```
[Install]
```

```
WantedBy=multi-user.target
```

Restarting the service

- Reloading the daemon
`sudo systemctl daemon-reload`
`sudo systemctl enable clamav-realtime.service`
`sudo systemctl start clamav-realtime.service`
- Verify the status
`sudo systemctl status clamav-realtime.service`

Evaluating the service using EICAR file

- Downloading a harmless test file
wget <https://secure.eicar.org/eicar.com>
- Output
 - ✓ File is scanned, found a signature match.
 - ✓ Moved the file to /var/quarantine automatically
 - ✓ Event is logged on /var/log/clamav/realtime.log

To view the logs,

```
cat /var/log/clamav/realtime.log
```

To check the quarantine files,

```
ls -l /var/quarantine
```

Advantages of ClamAV

- Free and Open Sourced
- Cross-Platform Support: Compatible with Linux, Windows, and macOS, making it suitable for diverse computing environments.
- Automatic Signature Updates and Quarantine.
- Comprehensive Malware Detection: Detects a wide range of threats, including viruses, worms, Trojans, spyware, adware, rootkits, malicious scripts, and infected archives.
- Lightweight and Resource Efficient: Consumes minimal CPU and memory resources, making it suitable for desktops, servers, and virtual machines.

Disadvantages of ClamAV

- Signature-Based Detection: ClamAV primarily relies on malware signatures, making it less effective against newly emerging or unknown (zero-day) threats.
- Limited Advanced Threat Detection: It lacks advanced security features such as behavioural analysis, machine learning-based detection, exploit prevention, and ransomware protection that are commonly available in commercial antivirus solutions.
- Lower Detection Rate for Sophisticated Malware
- Performance Impact During Large Scans: Scanning large directories, compressed archives, or file servers can consume significant CPU, memory, and disk I/O resources, potentially affecting system performance.
- Requires Manual Configuration: Features such as real-time monitoring, automatic quarantine, notifications, and scheduled scans require additional configuration and scripting, increasing deployment complexity.

Conclusion

Thus, the described project implemented real-time malware detection and prevention in a Linux environment. The solution used ClamAV, FreshClam, the clamd daemon, inotify tools, and Systemd to monitor user directories, scan them for malware, and respond accordingly to detected threats.

The implemented solution was able to detect newly created or modified files and scan them with the clamAV daemon in real-time. Once infected files were detected, the system was able to quarantine them, create log entries. The operation of the malware detection system was tested using the EICAR test file, which allowed to ensure that the real-time monitoring, scanning, quarantine operation correctly without infecting the system.

Overall, the project achieved its goals as it created a reliable real-time malware protection system that could detect, quarantine, and log malicious files automatically. Moreover, the project proved that various open-source software can be combined to achieve an end-to-end security solution, which could become the basis for an integrated SIEM, email notification system, cloud-based threat intelligence system, and advanced malware analysis platform.